.

Diogo Castro

Curriculum Vitae

Summary

I'm a Software Engineer based in Porto, Portugal.

My professional journey began as a C# developer, but functional programming soon piqued my interest and led me to learn Scala, Haskell, and PureScript. More recently, I've also been using Rust for commercial projects.

I love building robust, reliable, and maintainable applications.

I also like teaching and I'm a big believer in "paying it forward". I've learned so much from many inspiring people, so I make it a point to share what I've learned with others.

I blog about FP and Haskell at https://diogocastro.com/blog.

Experience

April 2020-Present **Team Lead, Software Engineer**, *Serokell*, Remote.

Led the internal development team in developing and maintaining several open-source libraries and applications written in Haskell, such as xrefcheck and tzbot. Authored tztime.

Led the development of Haskell Certification, a platform for assessing and certifying a candidate's Haskell skills (now operated by the Haskell Foundation).

Developed a Rust application using Tauri for controlling Molecular Rotational Resonance (MRR) spectroscopy instruments, performing measurements, and visualising and analysing the measurement data.

Helped grow the Haskell and Rust ecosystems for the Tezos blockchain by developing a varied set of tools: a parser, typechecker and interpreter for the Michelson language, a REPL, a test framework for smart contracts, a client library for interacting with the chain, among others.

Contributed to Serokell's blog.

Led an effort to onboard new hires and interns.

+351 914424559 • dc@diogocastro.com
diogocastro.com • dcastro • diogocastro.com/blog

Nov 2017-April 2020 **Principal Software Engineer**, *SpotX*, Belfast, UK.

Senior Software Engineer, *SpotX*, Belfast, UK.
Developed RESTful web services in Scala, using the `cats/cats-effect` framework and Akka HTTP. Used Apache Kafka for publishing of events, and Prometheus/Grafana for monitoring.

Worked on a service that aimed to augment and enhance Apache Druid, a timeseries database.

Authored a Scala library for calculating the delta of any two values of a given type using Shapeless, a library for generic programming.

Taught a weekly internal Scala/FP course, with the goal of preparing our engineers to be productive in Scala whilst building an intuition of how to program with functions and equational reasoning.

Trained new hires and helped them bridge the gap between what they already knew and the world of FP in Scala.

May 2013-Nov 2017 **Senior Software Engineer**, *NantHealth*, Belfast, UK.

Software Engineer, *NantHealth*, Belfast, UK.
Responsible for designing, unit testing, implementing and deploying a variety of applications, such as:

- RESTful APIs, using ASP.NET Web API, ActiveMQ, Couchbase, and MS SQL Server;
- Front-end single-page applications, using AngularJS, TypeScript, LESS;
- Internal libraries written in C#;
- Routing of messages between applications using Java and Apache Camel.

I've also built some small internal tools to help streamline my coworkers' and my day-to-day activities, using Scala, Haskell and PureScript.

I helped organise a weekly brown-bag session during lunch hour, in which people talked about topics that interested them. I sometimes brought an exercise for people to solve in a language of their choice and share their solutions at the end.

Aug 2012-Mar 2013 **Researcher**, *Fraunhofer*, Porto, Portugal.
Developed the navigation module for an Android application, using both turn-by-turn and landmark-based approaches. Studied and compared the efficiency of these approaches in navigating older adults with mild dementia. Developed complex heuristics to evaluate landmark data retrieved from OpenStreetMap, and used the device's sensors (e.g., gyroscope, accelerometer) to locate and navigate the user.

Talks

Jan 2019 **The Haskell Epidemic**, *The Crystal Ball BASH, Belfast*.
A presentation about some of Haskell's most influential features and how Haskell has shaped the software engineering landscape.

Recording: https://youtu.be/nnoOF1HeAls

Slides: https://talks.diogocastro.com/the-haskell-epidemic/

Open Source

linear-locks https://hackage.haskell.org/package/linear-locks

Leverages Haskell's Linear Types to provide locking primitives that are statically guaranteed to be deadlock-free.

haskell-flatbuffers https://hackage.haskell.org/package/flatbuffers

Haskell implementation of FlatBuffers, a protocol for memory-efficient serialisation, originally designed by Google. It uses TemplateHaskell for generating code from a given schema, `megaparsec` for parsing schemas and `binary` for low-level bytestring manipulation.

lsp-xreferee https://github.com/dcastro/lsp-xreferee

LSP server written in Haskell for validating cross-references throughout a git repository.

tztime https://hackage.haskell.org/package/tztime

Haskell library for safely manipulating dates and time in a timezone-aware fashion, avoiding common pitfalls such as ambiguous or invalid datetimes.

tzbot https://github.com/serokell/tzbot

Slack bot that detects messages with references to some point in time and automatically converts them to the reader's timezone.

Education

2007-2013 **Master's in Informatics and Computing Engineering**, *Faculty of Engineering of the University of Porto*, Portugal.